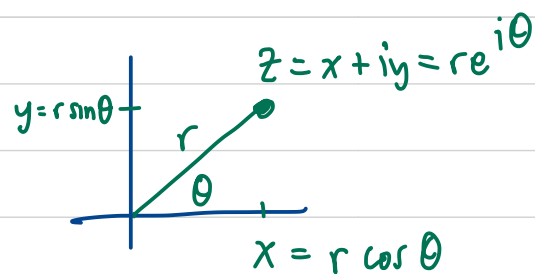


Chapter 0: preliminaries / vector spaces

complex numbers $\mathbb{C}$

$$i^2 = -1$$

$$z = x + iy$$



polar representation

length (modulus, magnitude): $r = |z| = \sqrt{x^2 + y^2}$

multiplication can be done algebraically or geometrically

$$(x + iy)(a + ib) = (xa - yb) + i(ya + xb)$$

$$(re^{i\theta})(se^{i\phi}) = \underbrace{rs}_{\substack{\text{lengths} \\ \text{multiply}}} e^{i(\underbrace{\theta + \phi}_{\text{angles add}})}$$

$$\text{Euler / de Moivre: } e^{i\theta} = \cos \theta + i \sin \theta$$

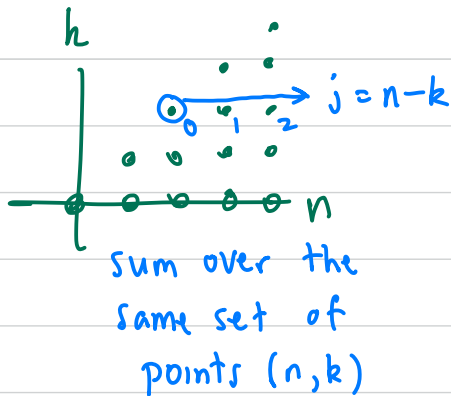
$$\begin{aligned} e^{i\theta} e^{i\phi} &= (\underbrace{\cos \theta \cos \phi - \sin \theta \sin \phi}_{\cos(\theta + \phi)} + i(\underbrace{\sin \theta \cos \phi + \cos \theta \sin \phi}_{\sin(\theta + \phi)}) \\ &= e^{i(\theta + \phi)} \end{aligned}$$

trig. addition formulas

$$\text{For } z \in \mathbb{C}, e^z \text{ is defined by } e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$$e^{(z+w)} = \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \frac{n!}{k!(n-k)!} z^k w^{n-k}$$

$\uparrow$ binomial theorem $\uparrow$ $j = n-k$



$$= \sum_{k=0}^{\infty} \frac{z^k}{k!} \sum_{j=0}^{\infty} \frac{w^j}{j!}$$

$$= e^z e^w$$

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} = \frac{1}{0!} + i \frac{\theta}{1!} + i^2 \frac{\theta^2}{2!} + i^3 \frac{\theta^3}{3!} + i^4 \frac{\theta^4}{4!} + \dots$$

$$= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right)$$

$$= \cos \theta + i \sin \theta$$

complex conjugate: $\bar{z} = x - iy$

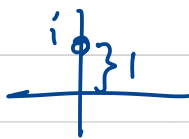
$$z \bar{z} = (x + iy)(x - iy) = (x^2 - (-y^2)) + i \underbrace{(yx - xy)}_0$$

$$= x^2 + y^2 = |z|^2$$

division: rationalize the denominator

$$\frac{z}{w} = \frac{z \bar{w}}{w \bar{w}} = \frac{z \bar{w}}{|w|^2} \leftarrow \text{real denominator}$$

$$\frac{x+iy}{a+ib} = \frac{(x+iy)(a-ib)}{(a+ib)(a-ib)} = \frac{xa+yb}{a^2+b^2} + i \frac{ya-xb}{a^2+b^2}$$

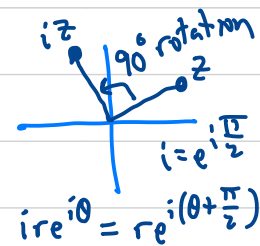
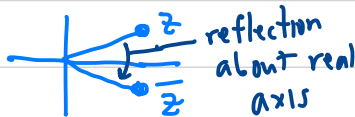
$$\frac{1}{i} = \frac{-i}{|i|^2} = -i$$


Real and imaginary parts

$$\operatorname{Re} z = \frac{z + \bar{z}}{2} = \frac{\cancel{(x+iy)} + \cancel{(x-iy)}}{2} = \frac{2x}{2} = x \quad \checkmark$$

$$\operatorname{Im} z = \frac{z - \bar{z}}{2i} = \frac{\cancel{(x+iy)} - \cancel{(x-iy)}}{2i} = \frac{2iy}{2i} = y \quad \checkmark$$

$\bar{z}$ vs iz :



Complex conjugation passes through most operations :

$$\overline{zw} = \bar{z} \bar{w}, \quad \overline{(z/w)} = \bar{z} / \bar{w}$$

If $f(z) = \sum_{n=0}^{\infty} C_n (z-a)^n$, C_n, a real

then $f(\bar{z}) = \overline{f(z)}$

example:

$$e^{\bar{z}} = \sum_n \frac{1}{n!} (\bar{z})^n = \sum_n \frac{1}{n!} \overline{z^n} = \overline{\sum_n \frac{1}{n!} z^n} = \overline{e^z}$$

we can also check this using

complex conjugation is continuous
(can take it outside of an infinite sum)

$$e^{\bar{z}} = e^{x-iy} = e^x e^{-iy} = e^x (\cos(-y) + i \sin(-y))$$

$$= e^x (\cos y - i \sin y) = \overline{e^x (\cos x + i \sin y)} = \overline{e^x e^{iy}}$$

$\uparrow$
 $\sin y$ is an odd function
 $\cos y$ is an even function

$$= \overline{e^{x+iy}} = \overline{e^z}$$

==

Vector spaces

Let K denote $\mathbb{R}$ or $\mathbb{C}$ (the field of scalars)

real numbers $\downarrow$
complex numbers $\downarrow$
terminology from abstract algebra $\downarrow$

A vector space V over K is a collection of objects
you can add together and multiply by scalars
(linear combinations make sense)

Examples: $\mathbb{C}^n$: n -tuples of complex numbers

$n=3$:

$$3 \begin{pmatrix} i \\ 2 \\ 0 \end{pmatrix} + i \begin{pmatrix} i \\ 0 \\ 1-i \end{pmatrix} = \begin{pmatrix} -1 + 3i \\ 6 \\ 1+i \end{pmatrix}$$

note that $\mathbb{C}^n$ can be regarded as a vector space over $\mathbb{R}$ of dimension $2n$. But you can only multiply by real scalars in that setting.

$M_{m \times n}$ = set of $m \times n$ matrices (over $\mathbb{R}$ or $\mathbb{C}$)

$m=3, n=2$: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} + 2 \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 5 & 4 \\ 5 & 8 \end{pmatrix}$ ← $\mathbb{R}$ case

$n=2$ similar to $\mathbb{C}^n$ regarded as a real vector space
 (first column = real parts)
 (2nd " = imaginary ")

The space of polynomials of $\deg \leq n$ is a vector space of dimension $n+1$:

$$p(z) = a_0 + a_1 z + \dots + a_n z^n$$

$$q(z) = b_0 + b_1 z + \dots + b_n z^n$$

$$(\underbrace{\alpha p + \beta q}_{\text{new polynomial}})(z) = \sum_{k=0}^n (\alpha a_k + \beta b_k) z^k$$

← coefficients combine just as in $\mathbb{C}^{n+1}$

$$\alpha \begin{pmatrix} a_0 \\ \vdots \\ a_n \end{pmatrix} + \beta \begin{pmatrix} b_0 \\ \vdots \\ b_n \end{pmatrix}$$

infinite dimensional spaces

$\ell^2(\mathbb{N})$ = space of infinite sequences $(x_1, x_2, x_3, \dots)$

such that $\sum_{n=1}^{\infty} |x_n|^2 < \infty \leftarrow$ square summable

$$\mathbb{N} = \{1, 2, 3, \dots\}, \quad \mathbb{N}_0 = \{0, 1, 2, 3, \dots\}, \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$\ell^1(\mathbb{Z})$ = doubly-infinite, summable sequences

$$(\dots, x_{-2}, x_{-1}, x_0, x_1, x_2, \dots)$$

$$\sum_{n=-\infty}^{\infty} |x_n| < \infty$$

(exercise done in class:
show that $\ell^1 \subseteq \ell^2$)

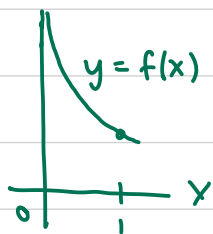
$L^1([a, b])$ = functions $f: [a, b] \rightarrow \mathbb{C}$ s.t.

$$\int_a^b |f(x)| dx < \infty$$

$L^2(\mathbb{R})$ = functions $f: \mathbb{R} \rightarrow \mathbb{C}$ s.t. $\int_{-\infty}^{\infty} |f(x)|^2 dx < \infty$

Example

$f(x) = \frac{1}{\sqrt{x}}$ is in $L^1((0, 1))$ but not $L^2((0, 1))$



$$\begin{aligned} \int_0^1 |f(x)| dx &= \left. \frac{x^{1/2}}{1/2} \right|_0^1 = 2 < \infty, & \int_0^1 |f(x)|^2 dx &= \int_0^1 \frac{1}{x} dx \\ &= \ln x \Big|_0^1 = \ln 1 - \underbrace{\ln 0^+}_{-\infty} = +\infty \end{aligned}$$

$L^4(\mathbb{R}^2)$ = functions $f: \mathbb{R}^2 \rightarrow \mathbb{C}$ s.t. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x, y)|^4 dx dy < \infty$